WEEK 8 · LECTURE L1 OF 3 · UNIT III

Dashboard Design

What an executive should see in ten seconds, and what to leave out.

Part 1 The ten-second test: hierarchy, restraint, the right chart

Part 2 Dashboard theater and the number that drives the wrong behavior

Part 3 Critique a sample GreenLeaf dashboard

Read Caldwell (2022), dashboards and data communication

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PART 1 · THE TEN-SECOND TEST

Can an executive grasp the message in ten seconds?

Hierarchy

The most important number is largest and first.

Restraint

Few tiles, little decoration, nothing without a decision behind it.

The right chart

Lines for change over time; bars for comparison.

Dashboard type	For whom	What it shows
Strategic	Executives	A few KPIs, monthly or quarterly
Operational	Managers	Daily work, updated often
Analytical	Analysts	Data to explore and filter

RULE Do not mix types on one screen. A CHRO screen full of filters is an analyst's tool.

PART 2 · DASHBOARD THEATER

Built to look rigorous, not to drive a decision

Target	Counter metric that reveals gaming
Chatbot deflection	Resolution rate
Time to fill	Quality of hire
Course completion	Assessment scores

WHY A number that becomes a target changes behavior. Pair every target with a counter metric that would expose gaming.

PART 3 · CRITIQUE

Five candidate tiles from GreenLeaf data

Tile	Type	Now	Counter metric
Overall attrition	Lagging	20.6%	Regretted exits among high performers
Staff under weak managers	Leading	14.3%	Exit rate in that group (37.1%)
Active self-service use	Leading	44.8%	Helpdesk tickets per employee
Tier 3 digital skills completion	Leading	39.4%	Assessment scores
Hiring adverse impact ratio	Compliance	0.60	Time to fill

YOUR TURN Which tile would you remove, and which decision does each remaining tile serve?

Going deeper: the theory underneath dashboard design

The next slides set out Few's definition, Tufte's data-ink ratio, Cleveland and McGill on graphical perception, and the limits of working memory: what each claims, how it works, where it breaks, and what it means at GreenLeaf.

THEORY IN DEPTH · FEW (2006)

A dashboard is a single screen for monitoring

THE CORE IDEA

A visual display of the most important information needed to meet objectives, on one screen, seen at a glance.

HOW IT WORKS

Four rules follow: most important only, one screen, glanceable, tied to objectives.

LIMITS AND CRITIQUE

Some argue interactive drill-downs make the single-screen rule outdated.

AT GREENLEAF If GreenLeaf's CHRO needs to scroll, the dashboard has already failed Few's test.

THEORY IN DEPTH · TUFTE (1983)

Maximize the data-ink ratio

THE CORE IDEA

Most ink on a chart should show data. Decoration that carries no data is chartjunk.

HOW IT WORKS

Erase non-data ink and redundant ink; keep only what informs.

LIMITS AND CRITIQUE

Studies show some embellishment can aid memory; minimalism is not always best.

AT GREENLEAF 3D pies and gradient tiles on an HR dashboard add ink and remove clarity.

THEORY IN DEPTH · CLEVELAND AND MCGILL (1984)

Some encodings are read more accurately than others

THE CORE IDEA

People judge position along a common scale best, then length, then angle, then area, then color.

HOW IT WORKS

Choose bar and dot charts for comparisons; avoid pies and bubbles where precision matters.

LIMITS AND CRITIQUE

Based on lab tasks; real dashboards also need speed and memorability.

AT GREENLEAF Show attrition by tier as sorted bars, not as a pie.

THEORY IN DEPTH · COWAN (2001); MILLER (1956)

Working memory holds very little

THE CORE IDEA

Miller suggested about seven items; Cowan's later review put the limit nearer four chunks.

HOW IT WORKS

Beyond a few items, viewers stop comparing and start skimming.

LIMITS AND CRITIQUE

Chunking and familiarity raise effective capacity.

AT GREENLEAF Five tiles is already near the limit. Fourteen tiles guarantee that nothing is compared.

TAKEAWAYS

Three things to carry out of this lecture

1

Pass the ten-second test: hierarchy, restraint, right chart.

2

Pair every target with a counter metric.

3

Fewer tiles, each tied to a decision.

NEXT Storytelling with data and the ethics of framing.